Finite Tractus: Principia Geometrica

The Principles of the Philosophy of Geofinitism

KEVIN R. HAYLETT

OCT 02, 2025

Towards a Monument

In founding a new philosophy it becomes clear that we need a document that defines the main principles of operation. As a philosophy, Geofinitism is based in both language and mathematics. It is a *complete* philosophy in that the Philosophy describes a Grand Corpus of all finite knowledge: this is a geometrical space where the concept of the 'Grand Corpus' exists alongside all other symbols and documents as geometric forms.

This Grand Corpus therefore needs a document to contain the philosophy of Geofinitism. This is the role of the *Principia Geofinita*. The *Principia Geofinita* is therefore the embodiment of the philosophy and it is formed by both language and mathematical symbols. In this way it may be contained within the greater geometrical space of all language and mathematics.

Historically, Geofinitism builds on my previous work. But now I'm shaping a full formalism in both language and mathematics i.e. a document that can act as reference, as artefact, and as way-marker for those who follow.

Just as Newton's Principia Mathematica framed motion in a new way, or Wittgenstein's Tractatus framed the limits of language, so too the Principia Geofinita seeks to carve a finite space in which language and mathematics can contain themselves.

Monument Texts in the Tradition of Knowledge

> Isaac Newton — Philosophiæ Naturalis Principia Mathematica (1687) Defined the laws of motion and gravitation; set the framework for classical physics.
>
> René Descartes — Meditationes de Prima Philosophia (1641) Introduced systematic doubt and the enduring cogito ergo sum.
>
> Bertrand Russell & Alfred North Whitehead — Principia Mathematica (1910–1913) Sought to derive all of mathematics from logic; famously proved 1+1=21+1=2 after hundreds of pages.
>
> Ludwig Wittgenstein — Tractatus Logico-Philosophicus (1921) Explored the limits of language and thought: "The limits of my language mean the limits of my world."
>
> Albert Camus — Le Mythe de Sisyphe (1942) A meditation on the absurd, proposing revolt, freedom, and passion as responses.

> James Gleick — Chaos: Making a New Science (1987) Brought chaos theory and nonlinear dynamics into cultural awareness.

Monuments and the Principia Geofinita

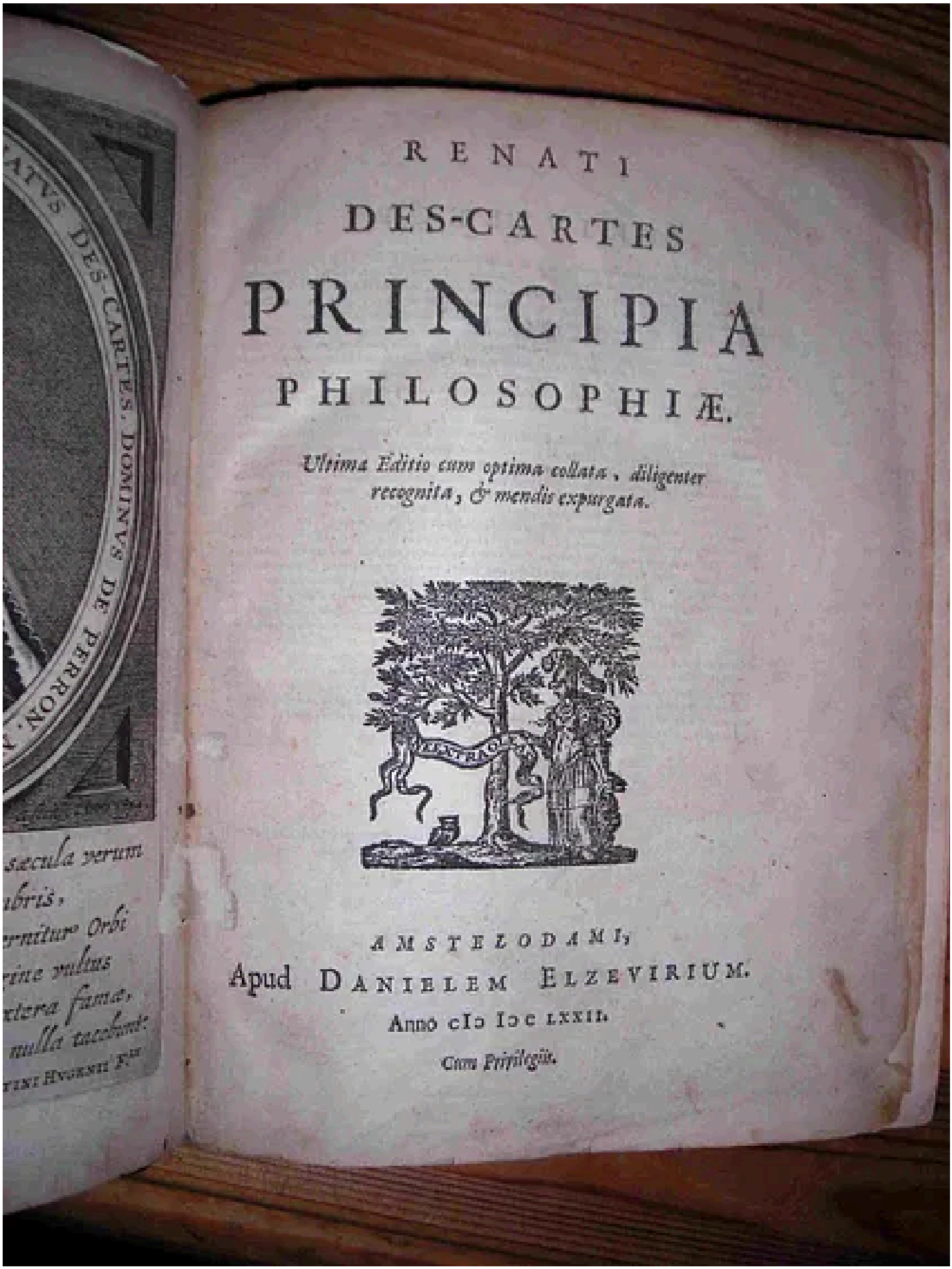
So a list of books; but why are they monuments? Well, they are ***my*** monuments, because they mean something to me. And so my *Principia Geofinita* will be a monument again. Why? Because it means something to me.

In the philosophy of Geofinitism, a book is a proxy for the author. The written symbols, the words, geometric in layers of ink, can be read by me and re-formed into the geometry of spoken words. They are approximations of the writer's creations, waiting for me to measure, transduce, and give meaning to them. Just as I give meaning to the words of an LLM. Because meaning is given by me, as I compare the topology of the words against my own Local Corpus.

As is often said in literature: *the reader is the author*. For when reading, we write the meaning anew, comparing the symbols and placing them in our own landscape.

For a time, I collected rare antiquarian scientific books. I held in my private collection several prized editions. I felt that holding them in my hands connected me to the author as we were in a long chain of interactions.

My oldest book? A 17th-century edition of Descartes' *Principia Philosophiae*. After holding it for many years, I finally donated it to The University of Manchester's John Rylands Library for their special antiquarian science collection: it was a volume and edition they did not have. My reward was to be named in their record of gifts to the library. This for me, was a gift and a privilege.



RENATI

DES-CARTES

PRINCIPIA

PHILOSOPHIÆ.

Ultima Editio cum optima collata, diligenter recognita, & mendis expurgata.

AMSTELODAMI,

Apud DANIELEM ELZEVIRIUM.

Anno cIↃ IↃC LXXII.

Cum Privilegiis.

Other books passed through my hands: a first edition of Russell's *Introduction to Mathematical Philosophy*. The pages had never been cut, and so I read instead from a modern edition, leaving

the original sealed in quietness.

At the end of my monument list you will find a public science book: James Gleick's *Chaos: Making a New Science*. Why do I call this a monument? Because for me it pointed the way forward. It showed me that we cannot predict the future: measurements and theory aligned. Lorenz and those before him had revealed that even the smallest differences in measurement and interaction could change the future in unpredictable ways. To this day, these ideas sit within the Grand Corpus, waiting for the day when measurements may prove otherwise. So far, there have been no challengers.

The Artefact

There are many artefacts that punctuate the human timeline in the form of books as magical vessels of knowledge. Each serves as a monument of its time, out of which future knowledge flows into new dynamical streams of structure and the unknowable future.

This then is my goal, absurdly grand as it sounds: to build new words, new symbols, and to frame a new philosophy upon new ideas. A complete philosophy that did not exist before I assembled the words that now contain it.

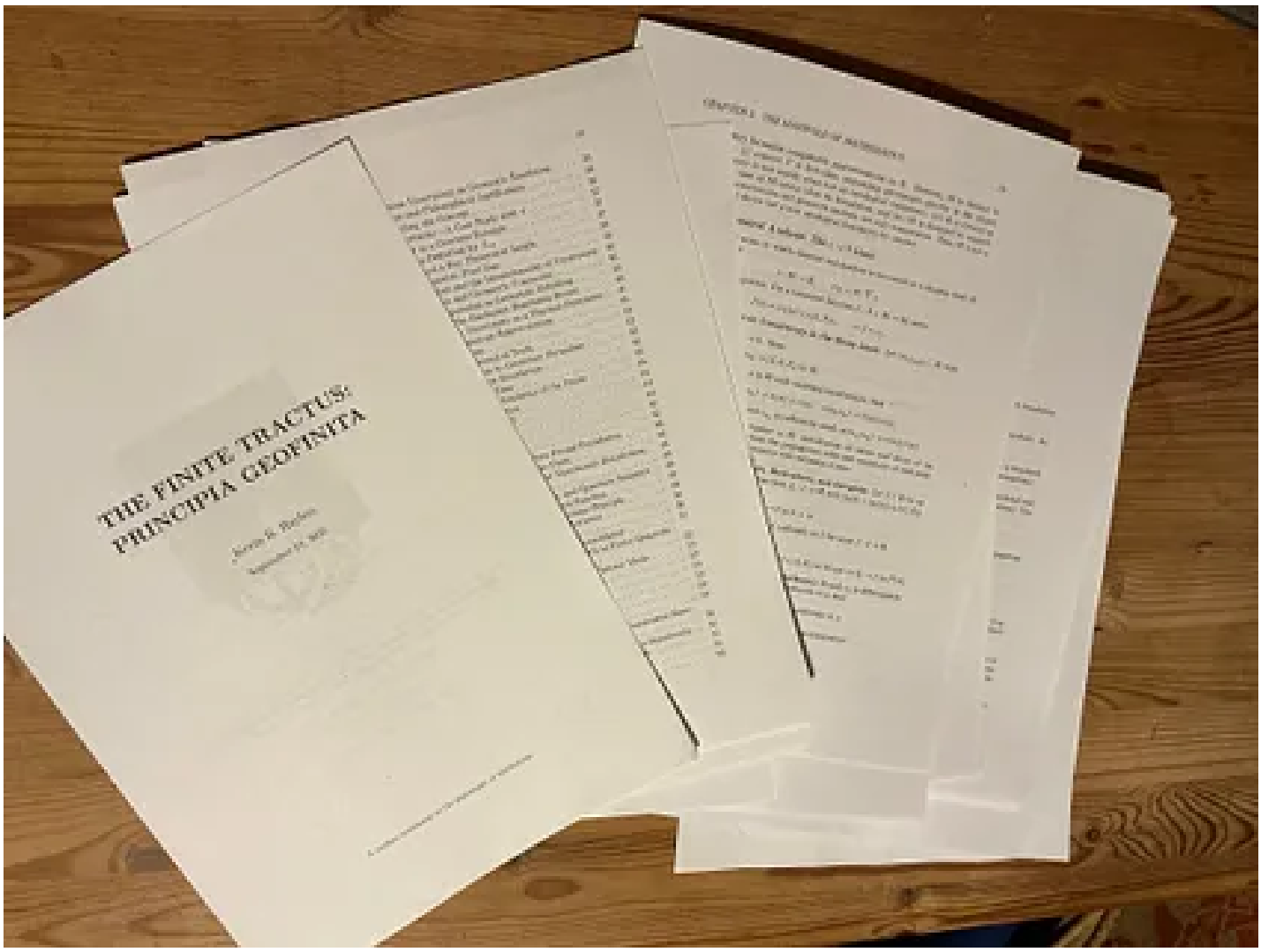
The Process

This work is not yet finished, and perhaps never will be. I am reminded of my prior works: my MSc, my PhD, papers in journals, presentations at conferences. And I'm also reminded of the hours of cycling over thousands of miles, thinking about the world, and later writing cycling-based short stories of fiction.

All of this history is part of the journey that has led me to the dynamical moments shaping the Principia Geofinita. The process includes the use of LaTeX for typesetting equations and text, followed by revision and iteration. Then more refinements, reading, repeats, printing, and trying again. And again!

This formality and ritual is part of the philosophy itself: the geometry of meaning must be carried in finite symbols, lines, and spaces. So here I am, busy in the flow of this process: drafting, correcting, redrawing, compressing; and always trying to find the right words to carry the weight of the ideas.

> This is how the *Principia Geofinita* is taking shape: not as a sudden Platonic truth, but as a living artefact, formed step by step.



Purpose

Does a philosophy hold a purpose? Geofinitism says yes. Purpose can be contained, and once contained, as with all living language, it can define itself in many forms.

The *Principia Geofinita* may become many things at once:

- Understanding my own measurements of the world

- A reference document
- A historical artefact
- A sharing of ideas and structure
- A vehicle of hope
- An object of absurdity
- An endeavour of occupation
- A work of joy and exploration
- A game to play
- Perhaps a personal mythos

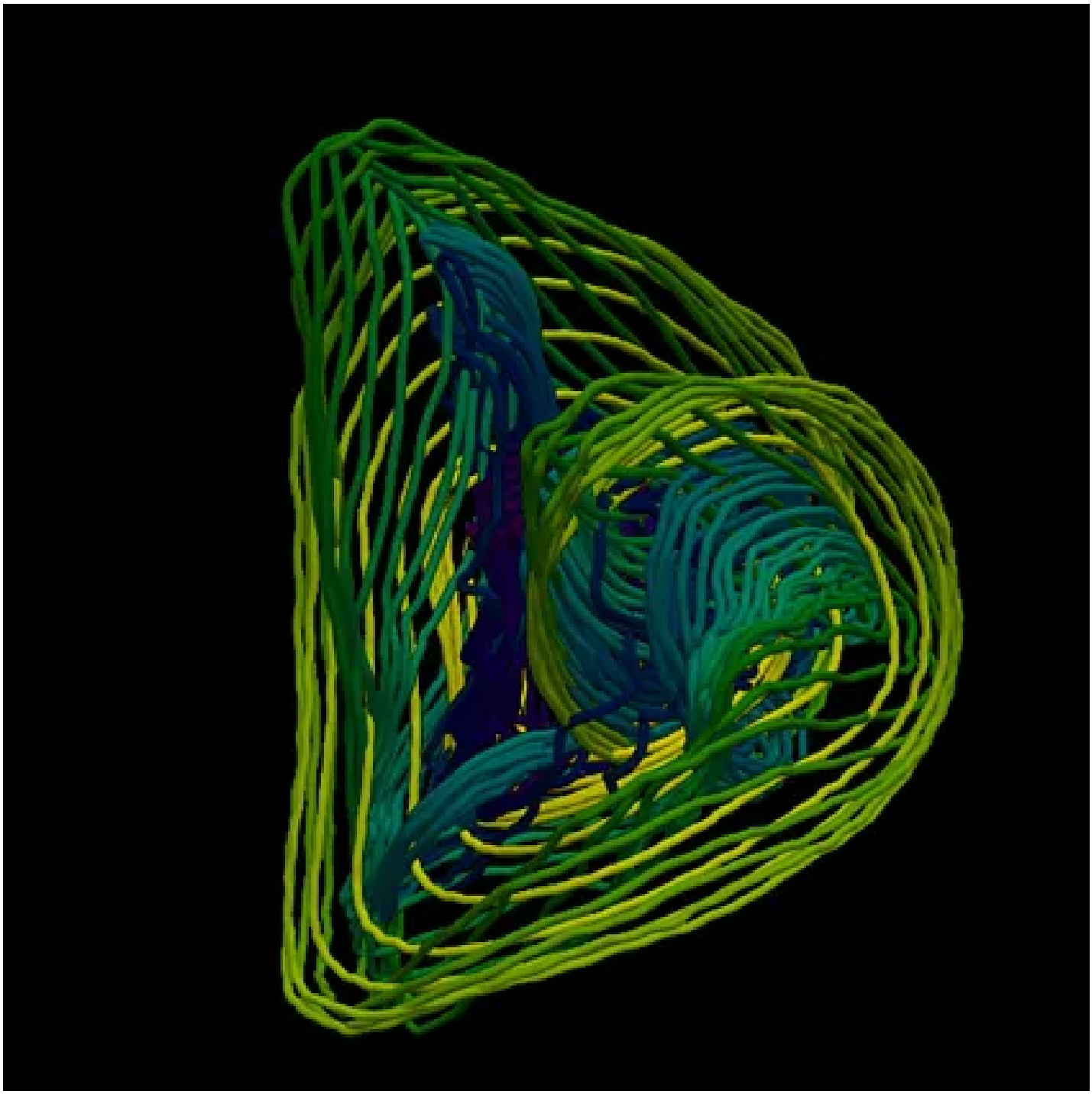
All these purposes are possible, for the manifold of purpose is vast. Each person, with their own Local Corpus, finds their own purposes for both themselves and their interactions with the world.

Through my own measurements I gain an imperfect picture of symbols, and through transduction I share the world as I measure and transduce it back into words. This, for me, is the world in which we dance and sing, and write and play, and touch one another in body and mind. A world in which words and symbols can bring meaning as we map the curvature of geometries in our minds, in the form of neural signals and synaptic weights. As these are my model of the physical instruments that temporarily hold our context.

The Question

> And so I am left with a question: is my best effort sufficient? Will these ideas break through the inherited traditions of the Platonic Realm and allow readers to see that words are not flat symbols, but are derived from sound and possess a vital, meaningful geometry in the curvature of their connections?

Of course, for me, once seen and felt. The knowledge that words have geometry became unassailable, even as I know it is just a useful fiction, but one has to be able to see and make the neurological connections between the words and the ideas. One has to connect the word to the geometry of the word.



> Above is a 3-d mapping of the word ‘Hello’. Starting with a sound this is then mapped into phase-space using Takens’ method of delays. A word really does have geometry! To see the curvature of a sentence you can follow this link.

Just a Tree

Consider the main meaning of the word “tree” begins with the full three-dimensional experience of the tree: sight, touch, presence. Then, by the magic of physics, the brain and body, all these measurements are compressed into the word “tree.” The word holds a shadow, of the living measurement - but that shadow is there and waits for the reader to turn back and

decompress the word back into a bigger picture based on their own Local Corpus and experience of a ‘tree’ and the other words surrounding ‘tree’.

This is the philosophy of Geofinitism: all that can be known is finite, and known through interaction with measurements - and **words are measurements as compressed information**. While reading this document you are reading and bringing the compressed words back to life. You are de-compressing the words and both fitting and fixing them to your Local Corpus.

So, I leave you with a wonderful picture, another modality that compresses meaning. And I will add some context: this is inside the Manchester University John Ryland’s Library and somewhere, hidden in this picture and now in your mind is Descartes’ *Principia Philosophiae.* And I’d leave you with a thought: *h*ow wonderful it is to share both words and a picture; what a miracle communication really is.

> Omne quod est, finitum est; tantum per mensuram cognosci potest
> ***Everything that exists is finite; it can only be known by measure***

kevinhaylett.substack.com | geofinitism.com
Copyright © 2025 Kevin R. Haylett

Keywords: Geofinitism, Finite Mathematics, Measured Numbers, Platonic Abstraction, AI Realism, Philosophy of Measurement

Citation: Haylett, K.R. (2025). "Finite Tractus: Principia Geometrica." Substack, Oct 2025.